

GREAT DUCK ISLAND, ON, Canada

WMO: 714620

Lat: 45.642N

Lon: 82.962W

Elev: 183

StdP: 99.15

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-19.8	-17.2	-24.7	0.4	-19.7	-22.1	0.5	-16.9	16.1	-6.3	14.6	-4.8	3.1	0	0.579

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	5.3	24.4	18.6	23.3	18.5	22.4	18.3	20.8	23.2	20.0	22.3	19.3	21.5	3.0	280

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.8	14.8	22.3	19.1	14.2	21.7	18.4	13.6	21.1	60.9	23.3	58.3	22.3	56.0	21.5	23.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
13.2	11.5	10.1	-23.6	26.5	3.4	1.6	-26.0	27.6	-28.0	28.6	-29.9	29.5	-32.4	30.6		
			DB	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	6.3	-6.2	-5.7	-2.2	2.9	8.2	13.6	18.2	19.0	16.0	9.8	3.2	-1.6
	DBStd	9.66	5.91	5.38	4.45	3.06	3.29	3.10	2.79	2.52	3.50	4.21	4.25	4.91
	HDD10.0	2236	504	439	377	214	75	4	0	0	3	56	205	359
	HDD18.3	4475	762	672	636	464	313	146	38	23	84	265	454	618
	CDD10.0	899	0	0	0	0	20	113	254	278	183	50	1	0
	CDD18.3	96	0	0	0	0	0	4	33	42	15	1	0	0
	CDH23.3	88	0	0	0	0	0	3	29	51	4	0	0	0
	CDH26.7	3	0	0	0	0	0	0	1	3	0	0	0	0
Wind	WSAvg	4.1	5.5	5.1	4.3	3.9	3.1	2.6	2.7	3.0	3.6	5.0	5.4	5.5
Precipitation	PrecAvg	822	54	40	46	65	65	85	72	89	89	83	76	59
	PrecMax	1228	108	77	126	140	125	296	119	182	205	188	206	109
	PrecMin	576	21	12	4	12	18	37	21	18	29	27	16	17
	PrecStd	145	20	17	30	32	28	53	32	36	38	39	42	24
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.2	4.7	7.7	11.6	19.7	23.2	25.6	26.4	23.9	20.5	12.3	9.0
		MCWB	4.2	3.4	4.1	6.6	13.3	16.8	20.7	18.2	19.7	16.8	10.4	8.2
	2%	DB	4.0	3.4	5.5	9.4	17.3	21.3	24.2	24.7	22.8	19.2	11.1	6.8
		MCWB	2.9	2.5	3.3	5.9	12.3	15.4	19.9	17.7	19.4	16.2	8.0	5.6
	5%	DB	2.8	2.5	4.2	7.8	15.2	19.8	23.1	23.6	21.8	17.7	10.0	5.6
		MCWB	1.8	1.5	2.5	4.8	11.8	14.9	19.0	17.9	18.8	14.8	7.4	4.3
	10%	DB	1.6	1.4	3.1	6.7	12.9	18.2	22.1	22.7	20.7	15.9	8.9	4.6
		MCWB	0.8	0.3	1.7	4.0	9.8	14.1	18.5	17.9	18.3	13.0	6.5	3.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.7	4.0	4.7	8.4	14.9	18.0	22.0	21.7	20.5	16.9	11.0	8.5
		MCDB	5.1	4.5	5.4	9.9	16.7	20.2	25.1	24.3	22.6	18.3	12.3	8.9
	2%	WB	3.1	2.7	3.5	6.7	13.1	16.2	20.9	20.8	19.8	15.9	10.0	6.2
		MCDB	3.7	3.4	4.4	8.1	15.5	18.5	23.5	23.3	21.6	17.3	11.2	7.2
	5%	WB	2.2	1.5	2.4	5.2	11.8	15.1	20.0	20.2	19.2	14.7	8.9	4.9
		MCDB	2.9	2.2	3.2	7.0	13.7	17.6	22.3	22.5	20.8	15.9	10.1	5.8
	10%	WB	1.2	0.2	1.6	4.2	10.4	14.2	19.1	19.5	18.5	13.5	7.5	3.6
		MCDB	2.1	1.0	2.5	6.0	12.1	16.6	21.6	21.7	20.1	15.1	8.7	4.7
Mean Daily Temperature Range	5% DB	MDBR	6.4	6.1	5.8	5.3	6.1	6.3	5.4	5.3	5.0	4.6	4.5	5.0
		MCDBR	5.3	4.8	5.7	7.0	8.2	7.3	6.1	5.9	5.1	3.6	4.3	4.6
		MCWBR	4.3	4.9	4.5	4.6	6.2	5.1	3.9	3.9	3.6	4.0	3.9	4.8
	5% WB	MCDBR	4.3	5.5	4.7	5.9	8.6	7.2	6.1	5.7	4.8	5.0	5.0	5.0
		MCWBR	4.1	5.4	4.1	4.5	6.2	5.0	3.8	3.6	3.3	4.8	4.5	5.1
Clear-Sky Solar Irradiance	taub	0.292	0.298	0.313	0.355	0.373	0.397	0.400	0.389	0.366	0.345	0.311	0.288	
	taud	2.440	2.387	2.407	2.346	2.400	2.357	2.376	2.404	2.453	2.476	2.517	2.497	
	Ebn at Noon	821	885	920	904	895	871	863	860	852	820	794	791	
	Edh at Noon	70	88	100	117	115	120	117	109	96	81	65	60	
All-Sky Solar Radiation	RadAvg	1.20	2.19	3.70	4.68	5.57	6.05	6.09	5.17	3.99	2.37	1.26	0.86	
	RadStd	0.13	0.21	0.33	0.44	0.52	0.48	0.24	0.27	0.36	0.25	0.18	0.09	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (31 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page